SWEEP rule on the KD-Toric code

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1 SWEEP Rule Decoder on the Toric

The decoder works as follows: each vertex v looks for a suggestion $\hat{v}$ on $\uparrow(v) \cap lk_2$ that matches σ_v^+. If there are more than one, then pick the one which has the minimal weight.

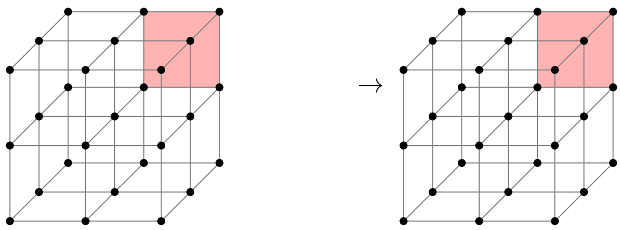

Figure 1: SWEEP rule

2 Span Expansion

Define the potentials functions $L_i : \mathbb{R}^d \to \mathbb{R}$ as follow[1]:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d+1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d+1 \end{cases}$$

The **Size** of a subset S is defined as:

$$\mathbf{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

Numerically, we show that the span property holds for the checks. Namely, let e be an unsatisfied check at time t, and let H_e be the unsatisfied checks at time $t-1$. Then there is a set $H'_e \subset H_e$ such that:

1. For any $i < d+1$ there exists $e' \in H'_e$ such $L_i(e') \geq L_i(e)$.
2. There exists $e' \in H'_e$ such that $L_{d+1}(e') \leq L_{d+1}(e) + 1$.

Proof. $\square$

[1]Notice that we use opposite signs relative to Berman and Peter Gacs, yet we agree with Perskil.